



Long-term care facilities and migration of elderly households in an aged society: Empirical analysis based on micro data

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ABSTRACT

With a simultaneous increase in long-term care (LTC) facilities, elderly households in Japan are increasing with geographic dispersion. In this study, using the household panel survey, we examined the relationship between LTC facilities and the migration, particularly moving out, of elderly households from their respective houses and municipalities. The results show that house owners do not typically move out from their housing. Meanwhile, renters tend to stay in municipalities where LTC facilities are accessible. In particular, if the number of beds at LTC facilities per hundred elderly increases by 10 percentage points, the probability of the elderly to move decreases by about 0.35–0.37 percentage points. In municipalities where the new type of elderly facilities was introduced in 2011, owner households headed by people aged 60–65 years and renters headed by aged 60–65 years and 75 years and above reduced their residential mobility.

1. Introduction

The rapid progression of aging populations and the weakening of the family's role as the primary caretakers of the elderly are common problems in developed countries. Japan is rapidly becoming one of the most aged countries in the world; its elderly ratio was 27.7% in 2017 (Cabinet Office, 2018). In addition to the high aging ratio, more than half of the households with elderly people consist of a single person or couples without children (Cabinet Office, 2018; Fig. 1-1-8). It may be difficult for members of such households to expect informal care from their family. Therefore, a long-term care (LTC) facility is an option when LTC is required.¹ According to a survey of people aged 55 years and above by the Cabinet office, 42.9% of men and 55.1% of women want to move to a location with LTC or medical facilities when they need LTC (Cabinet Office 2007). Therefore, as the number of single elderly households increases, the demand for LTC facilities will increase. However, there is a shortage of LTC facilities in Japan, and across

municipalities, many elderly people are awaiting admission.²

Since 2011, through mergers, the number of hospitals and clinics has decreased, leading to a more efficient use of medical resources. However, the number of beds at LTC facilities has increased. Therefore, the availability of LTC facilities has become one of the most important factors in defining the attractiveness of a municipality from the perspective of the elderly.

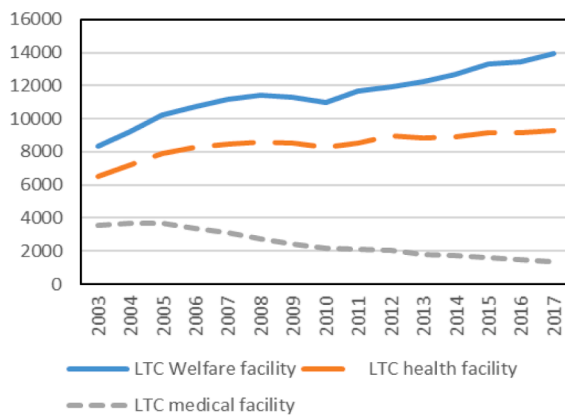
In such a situation, single elderly households may move to municipalities with sufficient LTC facilities when they require LTC. Inter-jurisdictional migration of the elderly has a significant influence on the demographics, fiscal performance, and social security policy of the concerned municipalities, as well as national social security policy. As emphasized by Newbold (2015), migration of the elderly is an important research subject in regional science. However, there is a lack of research on the mobility of elderly households in Japan. Moreover, most previous studies use aggregated data, and only a few studies use micro-level data. Starting from Nakazawa (2007), a series of studies based on Japanese

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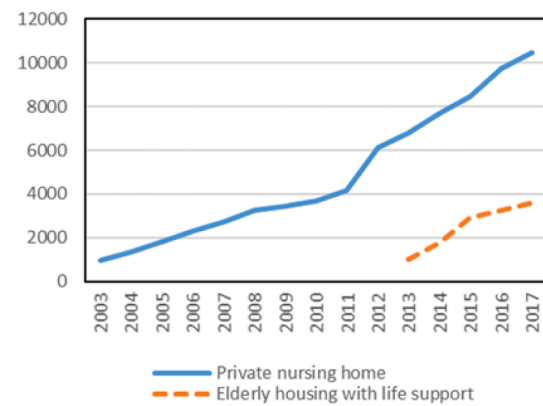
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¹ According to Hanaoka and Norton (2008) and Bonsang (2009), informal care by children and formal nursing care are substitutes for each other.

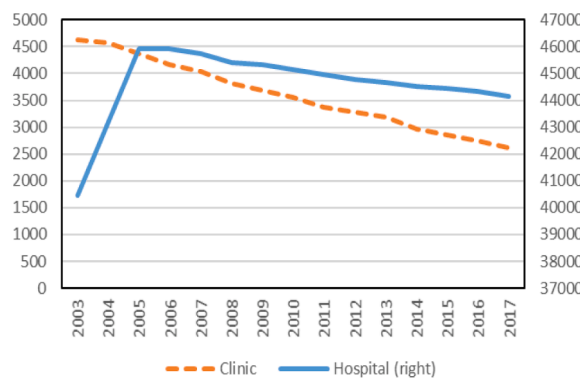
² According to an investigation by the Ministry of Health, Labour, and Welfare in 2006, the number of applicants waiting for admission to LTC facilities was about 385,500. The national average ratio of the number of those waiting for admission to existing facilities is 101%. Specifically, a large number of applicants waiting for admission reside in major cities. For example, the ratio is 126% in Tokyo and Kanagawa Prefecture.



(a) LTC facilities



(b) Private nursing homes and elderly housing with life support service



(c) Hospital and Clinic

Fig. 1. Average number of beds of long-term care facilities in prefectures: 2003–2017, (a) LTC facilities (b) Private nursing homes and elderly housing with life support service, (c) Hospitals and Clinics. Sources: (a) Ministry of Health, Labour, and Welfare, 2003–2017a, (b) Ministry of Health, Labour, and Welfare, 2003–2017b, (c) Ministry of Health, Labour, and Welfare, 2003–2017c.

municipal-level aggregated data has examined the relationship between the relocation of elderly households and LTC facilities, finding that the elderly moved to municipalities where LTC facilities were adequate.³

These studies use the net number of elderly movers at the municipal level over a 5-year period; they do not include the attributes and characteristics of these movers. By contrast, *Kojima (2013)* is the only study that uses Japanese household-level datasets. He examined the mobility of elderly households using the 7th National Survey on Migration in 2011. However, the study lacks variables related to economic factors such as household income and assets. In addition, the study does not distinguish housing tenure, and pools owner-occupied and rental housing samples together. Moreover, it ignores LTC facility variables. The parameter estimates derived by omitting important variables and ignoring differences in tenure may not be reliable. Therefore, to the best of our knowledge, there are no studies of elderly migration based on economics using micro-level data in Japan.

³ While *Nakazawa (2007)* relied on data for the Tokyo metropolitan area from 2000–2005 and *Nakazawa (2017)* expanded the dataset to 2005–2010, *Kawase and Nakazawa (2009)* and *Nakazawa and Kawase (2011)* expanded the analysis by *Nakazawa (2007)* to the national level for the same period and reached similar conclusions.

However, if we examine previous research based on micro-data from an international perspective, we find an abundance of studies on elderly migration and the determinants of mobility, such as separation from or death of a spouse (*Feinstein and McFadden, 1989; Venti and Wise, 1989; Angelini and LaFerrere, 2012*), retirement (*Ermisch and Jenkins, 1999*), geographic distribution (*Banks et al., 2012*), housing price volatility (*Banks et al., 2010*), property tax levels (*Shan, 2010*), and distance from children's residence (*Painter and Lee, 2009; Lee and Painter, 2014*). However, there are no studies that examine the availability of long-term care facilities as determinants of the migration decision of the elderly.

This study analyzes the mobility of elderly households, focusing on them moving out of present housing to understand the important determinants of mobility. It tests the hypothesis that elderly households with low ability of LTC provision within the household, such as those without children, move to municipalities with adequate LTC facilities. Moreover, we test this hypothesis based on the homeownership type (owner-occupied or renter-occupied) to show that the ease of relocation and the possibility of receiving family care affect the incentive to move to an LTC facility.

To test the hypothesis, we estimate models of elderly household mobility using a household panel survey, and data on municipality-level LTC facilities from 2004 to 2018. We conducted the analysis based on a sample grouped by housing tenure due to their differences in mobility

rate and economic resources, such as income and assets, especially, housing equity. Our category of households with children does not distinguish between households living with and without children at the time of the survey. Households without children have never had or raised children. We conduct this because housing tenure and having at least one child may be related to the bequest motive and could influence the likelihood of needing LTC when someone becomes frail or disabled.⁴

Moreover, we analyze a causal impact of the new type of elderly facility on residential mobility of the elderly. In November 2011, new type of long-term care facility called *elderly housing with life support service* was introduced. We consider that this is a natural experiment for households. The staggered difference in differences method (Athey and Imbens, 2018) is a useful statistical approach for investigating the effect of the introduction of the new mode of LTC facility on the mobility of elderly households as the number of such housing increased gradually since the introduction.

We obtain the following results. Owners do not move out from their houses, whereas renters tend to stay in cities where LTC welfare and private nursing homes are accessible. In particular, a 10 percentage point increase in the number of beds of facilities per hundred elderly decreases their probability to move by 0.35–0.37 percentage points. Our baseline models are robust even if we estimate the models using logit models considering the rare case of the elderly's mobility. From the second analysis, in the municipality that introduced elderly housing with life support service, owners, headed by people aged 60–65 years, reduce their probability to move by about 0.01 percentage points. Conversely, renters, headed by people aged 60–65 years and 75 years and above, reduce their residential mobility by about 0.06 and 0.08 percentage points, respectively.

The remainder of this paper is organized as follows. Section 2 briefly introduces the LTC system in Japan. Section 3 presents a simple model and Section 4 examines the data used in this study. Section 5 discusses the estimation results, while Section 6 presents our conclusions. Details of the data and models are explained in the Appendix.

2. Background of long-term care facilities in Japan

Japan's LTC system experienced a drastic change following the introduction of the public LTC insurance program in fiscal year 2000 (Campbell and Ikegami, 2000; Tsutsui and Muramatsu, 2005, 2007; Olivares-Tirado and Tamiya, 2014). The scheme widened the range and number of care receivers and provides coverage for both at-home services (at-home care) and services at facilities (institutional care). LTC insurance covers people older than 65 who need LTC services and those diagnosed with specific diseases. To use the services, the insured person needs to be certified as “requiring long-term care” by the local municipality based on an assessment of the amount of care required.⁵ Those who are insured can use LTC care from any provider, subject to a 10% self-burden of officially fixed prices specific to each type of care service. However, there is a limit to the amount of care that is covered by the insurance scheme.

Institutional care services are offered by the following facilities (Olivares-Tirado and Tamiya, 2014): (1) LTC welfare facilities for the elderly (special nursing homes for the elderly); (2) LTC health facilities for the elderly (geriatric health care facility); (3) LTC medical facilities for the elderly (geriatric medical care facility); (4) private nursing homes; and (5) elderly housing with life support services.

The first type of facility is defined as one offering 24-hour services related to daily living needs with no limit on residency duration. According to the Ministry of Health, Labor, and Welfare, as of 2013, there

Table 1

Housing tenure transition of owner-occupied households.

	Owner households	%	Renter households	%
Stayer	17,123	99.24	2,217	94.66
Mover	131	0.76	125	5.34
Tenure				
to Owner	73	0.42	32	1.37
	(55.7)		(25.6)	
to Renter	30	0.17	74	3.16
	(22.9)		(59.2)	
Unknown	28	0.16	19	0.81
	(21.4)		(15.2)	
Location				
outside prefecture	17	0.20	19	1.61
	(13.0)		(15.2)	
inside prefecture	86	0.50	87	3.71
	(65.6)		(69.6)	
outside municipality	23	0.13	13	0.56
	(17.6)		(10.4)	
inside municipality	63	0.37	74	3.16
	(48.1)		(59.2)	
Unknown	28	0.16	19	0.81
	(21.4)		(15.2)	
Total	17,254	100	2,342	100

Note: Household with the head aged 60 years and above. The number in parentheses is a percentage ratio within the mover sample.

Source: JHPS/KHPS 2004–2018.

were 52,000 elderly people on the waitlist for such facilities.⁶ Regarding admission to LTC welfare facilities, it depends on an imperious need such as health condition and household situation. LTC welfare facilities can admit those living outside the municipality where the facility is located. Therefore, the moved elderly are not automatically placed lower on the waitlist. To reduce applications from elderly people not in need of institutionalized care, only those certified as care level 3 or above can apply per the April 2015 amendment of the Long-term Care Insurance Act.

The second type of facility is for rehabilitation and recovery from illness and injury. This facility provides nursing care, long-term care under the supervision of medical management, functional training and other necessary medical treatment, and assistance in daily living to its residents. This facility differs from LTC welfare facilities in that it does not provide lifelong nursing care.

The third type of facility also includes sanatorium-type wards as well as a ward for elderly patients with dementia, and hospitals with enhanced LTC service provisions. Medical care per se is not included in the LTC program but is instead offered under the national healthcare system. However, since 2011, the government has been phasing out this type of facility and they will be shut down by fiscal year 2020.

While only public or nonprofit social welfare corporations can operate LTC facilities, private companies can operate private nursing homes, the fourth type of facility. They can be grouped into several types depending on the services that they provide. In general, residents need to pay a deposit and rent to buy a right to live in the home. Although the amount of the deposit and rent to enter the private nursing home depend on the facility, they are generally more expensive than LTC welfare facilities. Admission of a private nursing home depends on the order of applications. Private nursing homes are not classified as long-term care facilities in the LTC insurance system, but they are classified as housing for the elderly with at-home LTC. As private nursing homes can provide lifelong nursing care, this type of nursing home could become an alternative to LTC welfare facilities.

With the shortage of LTC welfare facilities and high upfront costs of private nursing homes, elderly housing with life support services was

⁴ Precise analysis of bequest motive and mobility of household is conducted in Seko et al. (2019).

⁵ The conditions requiring long-term care range from mild to serious in a multistep approach, while the eligibility degree ranges across seven levels.

⁶ Nihon Keizai Shinbun, March 25, 2014.

introduced in 2011. This is a facility similar to private nursing homes, but here, tenants can lease at lower costs compared with private nursing homes. Barrier-free design is mandatory for the residences, and safety confirmation and life consultation services are provided. The majority of residents are elderly people who need only a low level of care or who do not have care requirements. This type of facility was established in accordance with the Act on Housing for the Elderly, which was amended in 2011, with the aim of enabling elderly singles or elderly couples to live at ease in familiar surroundings with low costs.

We hypothesize that the availability of institutional care facilities is important as a determinant of the residential mobility of elderly households. According to previous studies that use municipal-level aggregated data, LTC facilities generate strong magnetic effects on migration of the elderly. Specifically, LTC welfare facilities and private nursing homes are important as determinants of residential mobility of elderly households because both facilities can provide lifelong nursing care. To analyze the residential mobility of elderly households based on household-level micro-data, we chose LTC welfare facilities and private nursing homes as the explanatory variables.

Fig. 1 depicts the transition of the average number of beds at LTC facilities in cities from 2005 to 2017. The number of beds at LTC welfare facilities, LTC health facilities, and private nursing homes has increased since 2011. However, the number of LTC medical facilities, hospitals, and clinics has decreased due to mergers. According to Maeda (2015), the integration of medical functions into large hospitals progressed from 2003 to 2013 and, as a result, the number of small hospitals and clinics is decreasing. Overall, this consolidating trend may be contributing to a more efficient use of medical resources and may improve the quality of care but at the cost of convenient access for patients.

3. Model

In this section, we describe the theoretical model used to analyze the data. Following the previous literature (Molloy et al., 2017; Røed and Schöne, 2012), we use an approach to model migration as the outcome of a process by which each individual weighs the benefits and costs of living in a different location against those in a present location.

We define V_{ijt} as the indirect utility function of the i^{th} household at present municipality (e.g., ward, city, town, village) j at time t and V_{iht} as the indirect utility function of the same household at a neighboring municipality h where the household could move to. We assume that households decide on relocation based on the information of the previous year. If the utility at the new location is greater than the present location, such as in Eq. (1), the household will move:

$$V_{iht} (Y_{it-1}, W_{it-1}, X_{it-1}, RP_{h,t-1}, A_{h,t-1}) > V_{ijt} (Y_{it-1}, W_{it-1}, X_{it-1}, RP_{j,t-1}, A_{j,t-1}). \quad (1)$$

Here, Y is the real household income, W is the real household wealth, X is the vector of the household's characteristics, RP_j is the ratio of rental and housing prices at the present municipality j , RP_h is the same ratio at the new location where the household moves to, A_j is the vector of the amenities at the present location, and A_h is the same vector at the neighboring location. We assumed that LTC institutions are included in this amenity.

The latent variable, M_{ijt}^* , as the difference of the utility and the specific function is defined as follows:

$$M_{ijt}^* = V_{iht} - V_{ijt} = V(Y_{it-1}, W_{it-1}, X_{it-1}, \Delta RP_{ijt-1}, A_{ih,t-1}, A_{ij,t-1}). \quad (2)$$

If $M_{ijt}^* > 0$, then the household will move out from the present residence and municipality; otherwise, the household will remain. Here, ΔRP is the difference in the rental price ratio, $\Delta RP = RP_h - RP_j$. For RP_h and A_h , assuming that households move within the larger region, we used the averages of the neighboring municipalities where the household can move to.

Based on these assumptions, we simply assume M_{ijt}^* as the following linear function:

$$M_{ijt}^* = \beta_1 Y_{it-1} + \beta_2 W_{it-1} + \beta_3 X'_{it-1} + \gamma_1 \Delta RP_{ijt-1} + \gamma_2 \ln A_{ih,t-1} + \gamma_3 \ln A_{ij,t-1} + u_{ijt}. \quad (3)$$

Further, we assume that mobility is diversely affected depending on natural logarithm of amenities of the neighboring municipalities, $\ln A_{ih,t-1}$, and the current municipality, $\ln A_{ij,t-1}$. The parameters of interest are γ_2 and γ_3 : γ_2 measures the pull force from neighboring municipalities, and γ_3 measures the push force from a present municipality. We mainly focused on γ_3 of LTC facilities to measure the force to move out from the current municipality. Positive values were obtained if the household moves out from the present municipality j , but negative values if the household remains in the current housing and municipality.

We use the following indicator for the latent variable M_{ijt}^* , which is 1 if the household moves and zero otherwise.

$$M_{ijt} = \begin{cases} 1 & : M_{ijt}^* > 0 \\ 0 & : \text{otherwise} \end{cases} \quad (4)$$

We assume that the error term of Eq. (3), u_{ijt} , is different depending on specifications. For example, in the linear models with fixed effects models, this is specified as follows: $u_{ijt} = v_i + \epsilon_{ijt}$, which includes fixed effects of households to catch the time-indifferent unobserved heterogeneities of households. In the logistic models, u_{ijt} is assumed to follow logistic distribution.

4. Data

4.1. Japan household panel survey

The “Japan Household Panel Survey” (JHPS/KHPS) is used to estimate the model. This is a combination of the former “Japan Household Panel Survey” (JHPS) and “Keio Household Panel Survey” (KHPS), which were previously conducted and managed as separate surveys.⁷ The characteristics of the surveys, such as the data structure and samples, are as follows.

The KHPS began in 2004 and surveyed 4005 households and the JHPS began in 2009 for 4022 households. In both surveys, households are selected through stratified two-stage sampling. Although the survey subjects of the KHPS include men and women aged 20–69 years in 2004 and those of the JHPS include men and women aged 20 years or above in 2009, the demographic characteristics of the survey responses are representative of Japanese households. Despite the overlap of the sampling populations, there is ultimately no overlap of KHPS and JHPS respondents. The two datasets have been combined since 2015 as they contain questions that are the same or similar. Our observation periods are from 2004 to 2018.

Drawing on JHPS/KHPS, we constructed a household sample limited to those headed by someone aged 60 years or older. Although an elderly household is defined in Japan's public statistics as one headed by an individual over 65 years old, our sample considers those with heads of households who are 60 years old or above as elderly households. This is because in many companies, a regular employee's mandatory retirement age is 60.⁸ Moreover, this lowering of the threshold age provides a larger sample.

We define the move of a household as the movement of a household to different housing over the last year. This definition includes migration, indicating moves within Japan over a distance to change

⁷ Panel Data Research Center at Keio University (2016)

⁸ According to a survey of the Ministry of Health, Labour, and Welfare (2016), the mandatory retirement age at 80.7% of companies is 60 years old.

municipalities and prefectures but excludes living in different housing in the same location. In this study, we mainly analyze moving out from the housing and municipality where they lived last year due to data limitations, described below.

Regarding the mover sample, attrition can be particularly serious as respondents who moved between the two waves are dropped from the sample, meaning a decrease in the mover sample. Furthermore, interviewees who moved to LTC facilities are included in the sample with attrition. This attrition is a limitation of measuring migration using panel data, which is mentioned in the previous literature (Kaplan and Schulhofer-Wohl, 2017). To overcome this problem, we used intensive follow up surveys from the sample to identify those households who were dropped but are confirmed to have moved, and included them as movers, and those confirmed to have stayed are included as stayers (Angelini and LaFerrere, 2012).⁹ Including these households, the mobility rate of owner-occupied households is 0.76 percent and that of renter households is 5.3 percent, including the households that dropped from the survey due to move-out during 2004–2018 (Table 1).¹⁰ Even if we do not know the households' destination, we can analyze the move-out of households from their present housing and municipalities. Therefore, the focus of our study is households' moving out from their current housing and municipality.

The overall numbers of households and observations in the survey are 10,465 and 76,016, respectively. Among them, the number of households headed by people aged 60 years and above and the corresponding observations are 3,697 and 19,596, respectively. We include the samples that have important variables with missing cases, such as income and assets, and use them by including the missing category dummy variables in the model.

4.2. Determinants of residential moves

We use three kinds of variables to explain the mobility of the elderly households: housing attributes, household attributes, and location. We use the number of rooms in the current residence for the scale of housing and dummy variables for the types of housing as variables for housing characteristics. For owned houses, we use the owner's self-reported current value of the house for the housing price. For renters, monthly rent is used to measure the cost of living.

For household attributes, we use the head of household's age to represent the life cycle of the household. Household income and savings, deflated by the consumer price index (2015=100, Tokyo), are used to capture household economic attributes. For the head of household's employment status, we used the dummy of the retirement of head of household because people tend to move when there is no longer a need to be near their workplace (Speare and Meyer, 1988). As elderly employment is increasing, we included a group of dummy variables for job status.

Dummy variables of marital status of head of household, number of household members, and number of children are used to represent the basic characteristics of the household. The number of children includes both members who reside in the same dwelling and those who do not. To represent households with a single occupant, we use a single member household dummy variable. These dummy variables represent the potential ability of a member within a household to take care of other members of the household. As residential mobility is often driven by

⁹ Interviewers must code the reason why they could not contact a household in the intensive follow up surveys. See Naoi (2007) regarding the attrition bias for the household residential mobility model using KHPS.

¹⁰ According to the latest "Housing and Land Survey of Japan" (2018), the total annual residential mobility rate was 3.5% between 2013 and 2018. If we focus on households headed by someone aged 60 years and above, the overall residential mobility rate is 0.4% moving from owner-occupied housing and 2.2% from rental housing, respectively.

demographic changes in households (Venti and Wise, 1989; Feinstein and McFadden, 1989), we use the dummy variables to represent households that experienced changes since the start of the survey in 2004. This variable includes a decrease of household members, due to death or moving away from their household members. Widowhood dummy is also included. This variable indicates death or separation of a spouse. As a proxy variable for health, self-rated health (SRH) is used (Chandola and Jenkinson, 2000). For SRH, we use the following question: "How is your health normally?" The respondent selects of the following choices: (1) Good, (2) Pretty good, (3) Normal, (4) Not so good, and (5) Bad. The dummy variable of university graduates among head of households is used to measure their permanent income.

Regarding location, because we cannot use the exact address of respondents and can only use the name of the municipality (ward, city, town, and village) where they reside, we use variables based on municipalities where respondents live. We use the municipality size dummy variables for households living in a large municipality and a medium-sized one to control for possible geographic differences in mobility behavior. The difference in the ratio of rental and housing prices of the location $\Delta RP = RP_h - RP_j$ is calculated from the rental and land price indices. For RP_h , the log of the ratio of rental to land price index of neighboring prefectures is used, and the log of the same ratio of the prefecture where the housing is located is used for RP_j .¹¹ The difference in these ratios is used to determine ΔRP . Eight regional dummy variables are used to represent the location of housing.¹² We used several variables for urban amenities such as number of parks, schools, climate, and so on, but these were not significant and showed a high correlation among explanatory variables in the preliminary estimation; we therefore used the variables related to LTC facilities.

For the amenity variable $A_{ih,t-1}$ of the neighboring municipality, where the household could move to, in Eq. (3), we calculated number of beds per hundred elderlies, $NBPE_{ih,t-1}^d$, for each LTC facility in the neighboring municipalities in d -km radius, where d is categorized into three levels: $d < 20$, $20 \leq d < 50$, and $d \geq 50$ km. Details for the variables are provided in Appendix A2. When we estimated Eq. (3) using $NBPE_{ih,t-1}^d$, we included the natural logarithm of variable of each LTC facility in the regression individually to prevent correlation between the facilities due to competition.¹³

We calculate the average number of beds of LTC and private facilities per hundred of elderly for each certain distance. We assume that the elderly, especially those who feel the need for care, are unlikely to choose a residence far from where they or their family members live. According to the Statistics Bureau of Japan (2013), 85% of the elderly households have children, and 80% of these elderly live within one hour of their children's houses. These elderlies may not want to sever ties with the community and their families even if they move to an LTC facility, and they are likely to have more information from nearby facilities than from distant facilities. The most important push (i.e., keeping elderly households in the living municipality or not) factor of the elderly

¹¹ As rent and price are measured as indexes r_t and p_t , respectively, the absolute level of the ratio is meaningless (Gallin, 2008, p. 641). Therefore, we re-specify as follows. We posit that R_B and P_B are the rent and price in the base year $t = B$, respectively, and r_t and p_t are the rent and price indices based on 1 at the base year $t = B$, respectively. As for the rents and prices at year t , $R_t = r_t \cdot R_B$ and $P_t = p_t \cdot P_B$. We take the logarithm of R_t/P_t : $\ln(R_t/P_t) = \ln((r_t \cdot R_B)/(p_t \cdot P_B)) = \ln(r_t/p_t) + \text{constant}$. Here, constant is $\ln(R_B/P_B)$. The log ratio of rent and price indices $\ln(r_t/p_t)$ captures the movement of the ratio of the rent and price of the location. We assume the constants of neighboring municipalities are the same and constants are assumed to be cancelled to capture differences with neighboring municipalities.

¹² Forty-seven prefectures are generally grouped into eight regions: Hokkaido, Tohoku, Kanto, Chubu, Kinki, Chugoku, Shikoku, and Kyushu/Okinawa.

¹³ We would like to thank the referee for pointing out the necessity to estimate the model by including each LTC facility in the regression individually.

Table 2
OLS estimation results of the linear probability models of residential mobility.

Facility/ Variables	1. Owner households		2. Renter households	
	(a) LTC welfare facility	(b) Private nursing home	(a) LTC welfare facility	(b) Private nursing home
Average no. of beds of facilities per hundred elderlies (0, 20) km, ln	−0.00180	0.00299	0.0219	0.00821
	(0.00301)	(0.00330)	(0.0239)	(0.0250)
Average no. of beds of facilities per hundred elderlies [20, 50) km, ln	0.00181	−0.00361	0.0156	−0.000201
	(0.00556)	(0.00476)	(0.0471)	(0.0357)
Average no. of beds of facilities per hundred elderlies [50 and above) km, ln	0.0150	0.00514	−0.0338	0.131
	(0.0489)	(0.0258)	(0.276)	(0.184)
No. of beds of facilities per hundred elderly, ln	−0.00239	0.00143	−0.0367*	−0.0347**
	(0.00271)	(0.00211)	(0.0214)	(0.0165)
Covariates				
Number of rooms [4 or over] (=1)	−0.00418	−0.00433	0.0120	0.0122
	(0.00417)	(0.00416)	(0.0111)	(0.0109)
Condominium (=1)	0.00831**	0.00832**	−0.0171	−0.0171
	(0.00421)	(0.00421)	(0.0138)	(0.0137)
Wooden apartment (=1)			0.00610	0.00701
			(0.0172)	(0.0171)
Ground lease (=1)	−0.00222	−0.00189		
	(0.00690)	(0.00686)		
Real housing equity, ln	−0.000740	−0.000740		
	(0.000509)	(0.000507)		
Real rent, ln			0.0150	0.0179
			(0.0127)	(0.0129)
Built after 1981 (=1)	−0.00244*	−0.00243*	−0.0169	−0.0160
	(0.00141)	(0.00141)	(0.0110)	(0.0110)
Age of household head (/10)	−0.0700*	−0.0703*	−0.00597	−0.0107
	(0.0373)	(0.0371)	(0.168)	(0.169)
Age of household head (squared, /100)	0.00485*	0.00487*	−0.000236	0.000160
	(0.00267)	(0.00266)	(0.0119)	(0.0120)
Real income, ln	0.000378	0.000339	0.00506	0.00401
	(0.00168)	(0.00168)	(0.00865)	(0.00867)
Real financial wealth, ln	−0.000180	−0.000181	0.00105	0.000845
	(0.000335)	(0.000336)	(0.00175)	(0.00176)
College graduate (=1)	0.00191	0.00192	0.0527***	0.0541***
	(0.00174)	(0.00171)	(0.0165)	(0.0166)
Married (=1)	−0.00227	−0.00227	−0.0133	−0.0122
	(0.00333)	(0.00334)	(0.0224)	(0.0220)
Female household head (=1)	0.00112	0.00118	0.00376	0.00387
	(0.00271)	(0.00270)	(0.0149)	(0.0147)
Single (=1)	0.00407	0.00399	0.0112	0.0123
	(0.00436)	(0.00434)	(0.0209)	(0.0207)
Self-rated health	−0.000311	−0.000298	−0.00477	−0.00462
	(0.000715)	(0.000719)	(0.00451)	(0.00454)
Full-time worker (=1)	0.00242	0.00250	−0.0140	−0.0146
	(0.00276)	(0.00275)	(0.0166)	(0.0167)
Part-time worker (=1)	0.000117	0.000136	0.0214	0.0211
	(0.00248)	(0.00248)	(0.0138)	(0.0139)
	−0.00226	−0.00217	−0.0227	−0.0201

Table 2 (continued)

Facility/ Variables	1. Owner households		2. Renter households	
	(a) LTC welfare facility	(b) Private nursing home	(a) LTC welfare facility	(b) Private nursing home
Self-employment (=1)				
	(0.00191)	(0.00192)	(0.0158)	(0.0158)
Retired (=1)	−0.00120	−0.00121	−0.00792	−0.00799
	(0.00167)	(0.00167)	(0.0134)	(0.0134)
Number of family members	0.000321	0.000349	0.00600	0.00432
	(0.000897)	(0.000893)	(0.0105)	(0.0103)
Family decreased since 2004 (=1)	0.00567**	0.00568**	−0.00896	−0.00797
	(0.00255)	(0.00255)	(0.0155)	(0.0154)
Widowhood (=1)	−0.00269	−0.00275	0.00531	0.00400
	(0.00468)	(0.00466)	(0.0188)	(0.0189)
Living with parents (=1)	−0.00202	−0.00210	0.00610	0.00799
	(0.00242)	(0.00243)	(0.0290)	(0.0287)
No. of coresident male children	−0.00146	−0.00145	0.0131	0.0173
	(0.00164)	(0.00164)	(0.0186)	(0.0186)
No. of non-coresident male children	−0.000110	−0.000109	0.0192	0.0200
	(0.00165)	(0.00165)	(0.0137)	(0.0139)
No. of coresident female children	−0.000309	−0.000401	−0.0181	−0.0162
	(0.00207)	(0.00207)	(0.0171)	(0.0171)
No. of non-coresident female children	−0.00110	−0.00110	−0.0149	−0.0150
	(0.00149)	(0.00148)	(0.0129)	(0.0130)
No child (=1)	−0.000641	−0.000716	−0.0180	−0.0180
	(0.00194)	(0.00193)	(0.0121)	(0.0120)
Rent-price rate and city size fixed effects	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Number of observations	17,254	17,254	2,342	2,342
R-squared	0.006	0.006	0.044	0.045
Number of events	131	131	125	125
Number of variables	60	60	60	60
EPV	2.183	2.183	2.083	2.083

Note: Cluster standard errors over households are in parentheses. Significance level: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. “km” and “ln” stand for the kilometers and natural logarithm, respectively. Coefficients of missing categories of number of rooms, rent, income, savings, and self-rated health are suppressed.

movement is the number of LTC beds in the municipality of the residence.

Moreover, we focus on the differences in the type of house ownership. The owner-occupied type of residence might be a disincentive for owners to choose institutional care when they require long-term care. The first reason is that owner-occupied houses are difficult to sell. The secondary market for detached houses is not well-developed and inactive (Kobayashi, 2016; Hashimoto et al., 2020). The average transaction rate of secondary housing is 14.5% in 2018, which is quite low compared to other developed countries (i.e., US: 81.0%, UK: 85.9%, and France: 69.8% in 2018; Ministry of Land, Infrastructure, Transport and Tourism, 2020). Housing buyers are difficult to find when you want to sell, especially, at depopulated suburbs. The second reason is the bequest motive for housing to their heirs is significant owing to the inheritance tax benefit of housing, and it decreases the probability to move to a facility (Seko et al., 2019). The third reason is the high possibility of at-home care. According to the Statistics Bureau of Japan (2013), 38% of

Table 3

Coefficients of the interaction term with LTC facilities in the current municipality and indicator of households' characteristics in linear probability model of residential mobility.

	1. Owner households		2. Renter households	
	(a) LTC welfare facility	(b) Private nursing home	(a) LTC welfare facility	(b) Private nursing home
(i) Age of household head (60,64)	−0.00179 (0.00393)	0.00279 (0.00410)	−0.0146 (0.0397)	0.00271 (0.0304)
(65,74)	−0.00365 (0.00359)	0.00143 (0.00249)	−0.0396 (0.0313)	−0.0349 (0.0175)
(75 and above)	0.00253 (0.00379)	0.00859 (0.00443)	−0.0798 (0.0295)	−0.0197 (0.0311)
(ii) Households without children	−0.00599 (0.0043)	0.00347 (0.0036)	−0.02291 (0.0379)	−0.02396 (0.0219)
(iii) Single male households without children	0.00095 (0.0113)	0.00088 (0.0123)	−0.05166 (0.0364)	−0.07259 (0.0397)
(iv) Single female households without children	−0.0462 (0.0230)	0.01022 (0.0169)	0.04803 (0.1296)	0.06421 (0.0557)
(v) Bad health condition	−0.00154 (0.0055)	0.03064 (0.0288)	−0.01977 (0.0247)	0.00129 (0.0271)

Note: Cluster standard errors over households are in parentheses. Significance level: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. The same covariates and fixed effects as Table 2 are used but their coefficients are suppressed.

Table 4

Staggered DID estimation of the model for moving out from owner-occupied housing.

Variables	1 Control 1	2 Control 1	3 Control 2 $d = 20$ km	4 Control 3 $d = 20$ km	5 Control 2 $d = 50$ km	6 Control 3 $d = 50$ km	7 Control 2 $d = 100$ km	8 Control 3 $d = 100$ km
Elderly housing (=1)	−0.0011 (0.00201)	0.00200 (0.00319)	0.000669 (0.00350)	0.00210 (0.00320)	0.00125 (0.00344)	0.00198 (0.00319)	0.00250 (0.00332)	0.00200 (0.00319)
Elderly housing × [55, 60](=1)		−0.00445 (0.00439)	−0.00792 (0.00494)	−0.00435 (0.00439)	−0.00796 (0.00488)	−0.00428 (0.00439)	−0.00849* (0.00487)	−0.00430 (0.00438)
Elderly housing × [60, 65](=1)		−0.00944** (0.00479)	−0.00735 (0.00489)	−0.0100** (0.00483)	−0.0102** (0.00508)	−0.0096** (0.00479)	−0.0107** (0.00507)	−0.0097** (0.00479)
Elderly housing × [65, 70](=1)		−0.00415 (0.00446)	−0.000507 (0.00444)	−0.00442 (0.00447)	−0.000740 (0.00446)	−0.00388 (0.00446)	−0.00201 (0.00441)	−0.00391 (0.00446)
Elderly housing × [70, 75](=1)		0.000360 (0.00446)	0.000130 (0.00442)	0.000245 (0.00448)	0.000151 (0.00433)	0.000507 (0.00446)	−0.000751 (0.00429)	0.000469 (0.00446)
Elderly housing × [75 and above] (=1)		−0.00569 (0.00423)	−0.00329 (0.00463)	−0.00571 (0.00425)	−0.00372 (0.00462)	−0.00558 (0.00423)	−0.00498 (0.00436)	−0.00562 (0.00423)
No. of observations	34,844	34,844	28,746	34,659	29,231	34,780	29,130	34,795
No. of households	6,200	6,200	5,735	6,181	5,795	6,193	5,799	6,193
Hausman stat: FE vs. RE	1,259	1,361	1,094	1,379	1,199	1,450	1,085	1,474
Hausman stat. d.f.	229	234	181	222	188	223	184	223
Hausman stat. P-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
F test for fixed effects	2.622	2.625	2.632	2.632	2.713	2.630	2.675	2.631
F test P value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Note: Cluster standard errors over households are in parentheses. Significance level: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. The same covariates as Table 2 are used, but their coefficients are suppressed. Households, region, municipality, and time fixed effects are included.

elderly owner-occupied households live with their children, and 54% live within 15 min of their children's houses. Therefore, they are more likely to receive public, at-home, long-term care, in addition to the support from their children when they need care. In addition, owner-occupied households are able to use public long-term care insurance services to make home renovating suitable for long-term care.

Meanwhile, elderly renter households can move out more easily than owner-occupied households. However, for renters, such renovation is difficult because renovating a rental housing requires negotiation with

the landlord and the house must be restored to its original condition when the tenant moves out. According to the [Statistics Bureau of Japan \(2013\)](#), 22% of elderly renter households live with their children, and 37% live within 15 min of their children's houses. In the case of elderly renter households, the likelihood of receiving care support from children living with or near them is lower than that of elderly owner households. Therefore, it is thought that elderly renter households have a higher incentive to move to a facility that provides long-term care services than elderly owner households.

Table 5

Falsification tests of staggered DID estimation of the model for moving out from owner-occupied housing.

Variables	1 Control 1	2 Control 1	3 Control 2 d = 20 km	4 Control 3 d = 20 km	5 Control 2 d = 50 km	6 Control 3 d = 50 km	7 Control 2 d = 100 km	8 Control 3 d = 100 km
Falsified elderly housing = Private nursing home								
Elderly housing (=1)	0.0057** (0.00285)	0.00893** (0.00370)	0.0127*** (0.00436)	0.00898** (0.00371)	0.0129*** (0.00426)	0.00884** (0.00370)	0.0127*** (0.00428)	0.00886** (0.00370)
Elderly housing × [55, 60](=1)		−0.00511 (0.00453)	−0.00834 (0.00531)	−0.00512 (0.00453)	−0.00896* (0.00522)	−0.00501 (0.00452)	−0.00880* (0.00525)	−0.00503 (0.00452)
Elderly housing × [60, 65](=1)		−0.00872* (0.00448)	−0.00837* (0.00498)	−0.0096** (0.00452)	−0.0104** (0.00504)	−0.0092** (0.00448)	−0.00991* (0.00508)	−0.00920** (0.00447)
Elderly housing × [65, 70](=1)		−0.00527 (0.00448)	−0.00385 (0.00467)	−0.00547 (0.00450)	−0.00435 (0.00476)	−0.00500 (0.00448)	−0.00488 (0.00492)	−0.00503 (0.00448)
Elderly housing × [70, 75](=1)		−0.000866 (0.00415)	−0.000723 (0.00444)	−0.000986 (0.00416)	−0.00201 (0.00416)	−0.000733 (0.00414)	−0.00198 (0.00420)	−0.000768 (0.00414)
Elderly housing × [75 and above](=1)		−0.00448 (0.00399)	−0.00466 (0.00395)	−0.00440 (0.00401)	−0.00686 (0.00447)	−0.00427 (0.00400)	−0.00689 (0.00437)	−0.00431 (0.00400)
No. of observations	34,844	34,844	29,305	34,660	29,753	34,780	29,670	34,795
No. of households	6,200	6,200	5,785	6,181	5,838	6,193	5,847	6,193
Hausman stat: FE vs. RE	1,326	1,348	1,194	1,388	1,239	1,448	1,072	1,479
Hausman stat. d.f.	229	235	187	222	187	223	184	223
Hausman stat. P-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
F test for fixed effects	2.626	2.625	2.636	2.633	2.731	2.631	2.682	2.632
F test P value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Note: Cluster standard errors over households are in parentheses. Significance level: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. The same covariates as Table 2 are used, but their coefficients are suppressed. Households, region, municipality, and time fixed effects are included.

Descriptive statistics of variables among owner-occupied households are tabulated in Table A 1 in the Appendix and those for renter households are in Table A 2.

5. Estimation results

First, to obtain the determinants of housing and household characteristics and amenities of the municipality on residential mobility, we estimated linear probability models of Eq. (3) by ordinary least squares (OLS) to start because of their simplicity and their coefficients approximate the partial effects of the explanatory variables (Wooldridge, 2010, pp. 562–565). However, there are several concerns about the results. We checked the robustness from the following three points: (i) estimation considering unobserved heterogeneity, (ii) estimation considering rare event nature of outcome, and (iii) estimation with alternative modeling approaches. Details are provided in Appendix A3.

5.1. Baseline results

Table 2 summarizes the estimation results of the models for movers for owner and renter households using capacity measures of LTC welfare facilities and private nursing homes in the current living municipalities and neighboring cities. As we have mentioned previously, to prevent correlation due to competition between LTC welfare facilities and private nursing homes, we included one variable of each facility in one regression.

Owners do not respond to the LTC facilities as these do not show significant results, whereas renters respond to LTC wealth facilities and private nursing homes that provide lifelong nursing care. A 10 percentage increase for the LTC welfare facility per hundred elderly in the living municipality decreases their probability to move by 0.367 percentage point, and the same increase for private nursing homes decreases their probability by 0.347 percentage points. These differences in the response of owners and renters to the capacity of the LTC facilities are consistent with the hypothesis we have mentioned in Section 4.2; that is, elderly renter households have a higher incentive to move to a facility that provides long-term care services than elderly owner

households. Meanwhile, distance-based LCT facility variables have no significant effect on both owners and renters.

Regarding household characteristics, coefficients of age of owner households show a downward convex implying that the probability to move decreases until a person reaches age 70 and, subsequently, increases. For renters, such relationships were not observed. From the economic variables, owners and renters are generally less sensitive to income and financial wealth. These are confirmed by the non-significant coefficients of real income and financial wealth. However, coefficients of real income have positive coefficients. This corresponds to the elderly mobility in European countries (Angelini and LaFerrere, 2012) and in the U.S. (Banks et al., 2012). However, from the insignificant coefficients, those effects are relatively weak compared to Western countries. Significantly positive coefficients of the college graduate dummy in renter's models indicate that renter households headed by college graduates, which imply higher permanent income, tend to move.

Since the survey started in 2004, owner households that experience family decline (except for widowhood) tend to move out. The widowhood dummy variable shows negative but insignificant results. The results suggest that owner households experiencing decline in members move out.¹⁴ However, households that experienced the death of, or separation from, a spouse tend to stay. The former result seems to correspond to Venti and Wise, 1989 and Feinstein and McFadden (1989). The latter result corresponds to Kojima (2013), who analyzed elderly Japanese households. This study suggests that, after the death of a spouse, the children's family move into their parent's house, and begin cohabitation. Meanwhile, in renter household models, both variables are not significant and the signs are opposite to those of owner models: the family decrease indicator is negative, and widowhood is positive.

The coefficients of the self-rated health index show negative and insignificant results in both owner and renter models. We did not find any significant results for self-rated health and residential mobility.

¹⁴ Death of a family member is the main reason and accounts for 81% and 84% of owner and renter samples, respectively.

Table 6

Staggered DID estimation of the model for moving out from rental housing.

Variables	1 Control 1	2 Control 1	3 Control 2 <i>d</i> = 20 km	4 Control 3 <i>d</i> = 20 km	5 Control 2 <i>d</i> = 50 km	6 Control 3 <i>d</i> = 50 km	7 Control 2 <i>d</i> = 100 km	8 Control 3 <i>d</i> = 100 km
Elderly housing (=1)	0.00298 (0.0123)	0.0253 (0.0182)	0.0261 (0.0229)	0.0283 (0.0182)	0.0147 (0.0232)	0.0253 (0.0182)	0.0140 (0.0234)	0.0253 (0.0182)
Elderly housing × [55, 60](=1)		−0.0559** (0.0266)	−0.0544* (0.0293)	−0.0586** (0.0265)	−0.0456 (0.0298)	−0.0559** (0.0266)	−0.0377 (0.0305)	−0.0559** (0.0266)
Elderly housing × [60, 65](=1)		−0.0559** (0.0247)	−0.0664** (0.0301)	−0.0589** (0.0248)	−0.0471 (0.0305)	−0.0559** (0.0247)	−0.0503 (0.0310)	−0.0559** (0.0247)
Elderly housing × [65, 70](=1)		−0.0244 (0.0251)	−0.0181 (0.0297)	−0.0265 (0.0251)	−0.0158 (0.0290)	−0.0244 (0.0251)	−0.00862 (0.0293)	−0.0244 (0.0251)
Elderly housing × [70, 75](=1)		0.0131 (0.0316)	0.0182 (0.0364)	0.0107 (0.0317)	0.0134 (0.0362)	0.0131 (0.0316)	0.0209 (0.0363)	0.0131 (0.0316)
Elderly housing × [75 and above] (=1)		−0.0808** (0.0355)	−0.0948** (0.0455)	−0.0821** (0.0354)	−0.0629 (0.0428)	−0.0808** (0.0355)	−0.0694 (0.0432)	−0.0808** (0.0355)
No. of observations	6,200	6,200	4,997	6,163	4,998	6,196	4,948	6,196
No. of households	1,408	1,408	1,274	1,400	1,289	1,404	1,276	1,404
Hausman stat: FE vs. RE	534.9	458.1	532.6	477	485.3	465.2	421.4	465.2
Hausman stat. d.f.	162	168	141	167	141	168	144	168
Hausman stat. P-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
F test for fixed effects	2.045	2.050	1.957	2.055	1.975	2.056	1.979	2.056
F test P value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Note: Cluster standard errors over households are in parentheses. Significance level: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. The same covariates as Table 2 are used, but their coefficients are suppressed. Households, region, municipality, and time fixed effects are included.

Table 7

Falsification tests of staggered DID estimation of the model for moving out from rental housing.

Variables	1 Control 1	2 Control 1	3 Control 2 <i>d</i> = 20 km	4 Control 3 <i>d</i> = 20 km	5 Control 2 <i>d</i> = 50 km	6 Control 3 <i>d</i> = 50 km	7 Control 2 <i>d</i> = 100 km	8 Control 3 <i>d</i> = 100 km
Falsified elderly housing = Private nursing home								
Elderly housing (=1)	−0.0183 (0.0200)	0.00700 (0.0233)	0.0229 (0.0489)	0.00940 (0.0232)	0.0141 (0.0484)	0.00700 (0.0233)	0.0101 (0.0489)	0.00700 (0.0233)
Elderly housing × [55, 60](=1)		−0.0628** (0.0250)	−0.0658** (0.0270)	−0.0652*** (0.0251)	−0.0649** (0.0273)	−0.0628** (0.0250)	−0.0545* (0.0284)	−0.063** (0.0250)
Elderly housing × [60, 65](=1)		−0.0432 (0.0307)	−0.0658* (0.0358)	−0.0456 (0.0308)	−0.0524 (0.0367)	−0.0432 (0.0307)	−0.0573 (0.0371)	−0.0432 (0.0307)
Elderly housing × [65, 70](=1)		−0.0663** (0.0267)	−0.0866*** (0.0331)	−0.0678** (0.0267)	−0.0769** (0.0340)	−0.0663** (0.0267)	−0.0635* (0.0334)	−0.066** (0.0267)
Elderly housing × [70, 75](=1)		0.00668 (0.0296)	0.00803 (0.0352)	0.00462 (0.0297)	0.00161 (0.0357)	0.00668 (0.0296)	0.00593 (0.0356)	0.00668 (0.0296)
Elderly housing × [75 and above] (=1)		−0.0780 (0.0531)	−0.113 (0.0835)	−0.0790 (0.0530)	−0.0549 (0.0726)	−0.0780 (0.0531)	−0.0583 (0.0774)	−0.0780 (0.0531)
No. of observations	6,200	6,200	5,106	6,163	5,117	6,196	5,078	6,196
No. of households	1,408	1,408	1,282	1,400	1,297	1,404	1,289	1,404
Hausman stat: FE vs. RE	466.8	486	480.1	495.6	460	480.9	431.1	480.9
Hausman stat. d.f.	162	166	146	167	146	167	148	167
Hausman stat. P-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
F test for fixed effects	2.049	2.055	1.961	2.059	1.962	2.060	1.958	2.060
F test P value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Note: Cluster standard errors over households are in parentheses. Significance level: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. The same covariates as Table 2 are used, but their coefficients are suppressed. Households, region, municipality, and time fixed effects are included.

The possibility of at-home family caregiving depends not only on whether the child lives with the elderly family but also on whether the child is providing care. Therefore, in this estimation, we considered not only whether the number of children lived with the elderly family but also the gender of the child. The results are not significant. Conversely, the points estimate of “No Child” show negative sign but are not significant.

5.2. Results of the models with interaction terms

The demand for LTC facilities may change with the characteristics of the households, such as the ability to provide nursing care inside the household, health conditions, and members' ages. To verify this case, we estimated the models with interaction terms of LTC facilities in the current living municipalities and variables related to household characteristics. We used the variable z_{it} to calculate the interaction terms

with variables of LTC facilities as follows: (i) age of head of households, (ii) an indicator of households without children, (iii) an indicator of male single households without children, (iv) an indicator of female single households without children, and (v) an indicator of bad health condition. The estimated model is expressed as follows:

$$M_{it} = \beta_1 Y_{it-1} + \beta_2 W_{it-1} + \beta_3 X'_{it-1} + \gamma_1 \Delta RP_{it-1} + \gamma_2 A_{ij\ t-1} + \gamma_3 A_{ih\ t-1} \times Z_{it-1} + \gamma_4 A_{ij\ t-1} + \gamma_5 A_{ij\ t-1} \times Z_{it-1} + \gamma_6 Z_{it-1} + u_{ijt}, \quad (5)$$

where Z includes a scalar of indicator or vector of indicators of households' characteristics. Given that we are interested in the moving out from the current municipality, estimates of $\partial \text{Prob}(M_{it} = 1) / \partial A_{ij\ t-1} = \gamma_4 + \gamma_5$ were tabulated in Table 3. To save space, only estimation results related to the estimates were reported.

Table 3 demonstrates the results of the model with interaction terms of LTC facilities and an indicator of households' characteristics. Among owner households, single female households without children respond to the LTC welfare facility. They reduce the probability to move by 0.46 percentage points if the LTC welfare facility per hundred elderlies in the living municipality increase by 10 percentage points.

For renter households, from the models with interaction terms of age of households' head, younger elderly aged 65–74 years respond to the private nursing homes, and older elderly aged 75 years and above respond to the LTC welfare facilities. These results are reasonable because private nursing home admit people aged 65 years and above irrespective of their care level and some renters may consider the home as a housing to live. Meanwhile, as LTC welfare facilities only admit people certified as care level 3 or above, older elderly aged 75 years and above may enter the facility to prepare for changes that might occur with the aging process. According to Ministry of Health, Labor and Welfare (2018), the ratio of the elderly aged 75 years and above account for 92.4% of people living at LTC welfare facilities. Single male households without children tend to respond to private nursing homes: 10 percentage point increase per hundred elderly in the living municipality decreases their probability to move by 0.73 percentage points.

5.3. Introduction of the elderly housing with life support service in 2011

As mentioned, in 2011 the Japanese government introduced a new type of long-term care facility called *elderly housing with life support service*. The purpose was to provide an LTC alternative because of a shortage of institutions and high upfront costs of private nursing homes. This provides us with an opportunity to examine the effect of this new type of facility on residential mobility of the elderly by using panel data.

In panel data, the staggered difference in differences method (staggered DID hereafter) (Athey and Imbens, 2018) is a useful statistical approach for investigating the effect of the introduction of a new mode of LTC facility on the mobility of elderly households as the number of such housing has increased gradually since the introduction. The ratio of respondents of JHPS who are living in municipalities with this new elderly housing increased from 40.6% in 2013 to 80.5% in 2017. Although the new housing was introduced in November 2011, municipalities only started to record statistics for it from 2013 (Ministry of Health, Labour, and Welfare, 2013–2018). Therefore, we dropped the data observed in 2012. In addition, we conducted the analysis for a sample of respondents aged 40 years and above to secure enough samples and examine the changes due to age, using the interaction terms with indicators of elderly households.¹⁵ Based on the data, we estimated the effect of the introduction of the new elderly

housing on residential mobility of elderly households.

For the treatment group, we used households living in municipalities that had elderly housing with life support services. When the elderly housing was introduced in 2015 in a municipality, a treatment indicator takes one in 2015 and keeps taking one in 2016 and 2017.

We developed three groups for the control groups:

Control 1: For the first group, we simply used the households living in municipalities where the elderly housing was not established.

Control 2: For the second group, we used households living in municipalities where the elderly housing was not established but where more LTC welfare facilities were accessible compared to neighboring municipalities within d km radius ($d = 20, 50, 100$).

Control 3: For the third group, we used households living in municipalities where the elderly housing was not established and LTC welfare facilities were not accessible compared to neighboring municipalities within d km radius ($d = 20, 50, 100$).

The move from the municipalities at $t + 1$ of the household i (M_{ijt}) can be studied using the staggered DID specification with treatment dummy (w_{jt}), which takes one if the elderly housing was established in the municipality j where household i lived at time t , and zero otherwise:

$$M_{ijt} = \alpha + \beta w_{jt-1} + \gamma X'_{it-1} + \mu_t + v_i + \xi_j + \epsilon_{ijt} \quad (6)$$

where X_{it} is a vector of covariates of households and housing characteristics and μ_t is time fixed effects, v_i is household fixed effects, ξ_j is municipality fixed effects, and ϵ_{ijt} is stochastic error term. β is a parameter of interest. If $\beta \leq 0$, the household in the treatment group stays in the present housing, $\beta > 0$ indicates the household moved out from the present housing.

Furthermore, to test the hypothesis that β changes depending on the head of households age, we multiply age group dummies H_{it-1} with w_{jt-1} :

$$M_{ijt} = \alpha + \beta w_{jt-1} + \delta H_{it-1} + \theta w_{jt-1} H_{it-1} + \gamma X'_{it-1} + \mu_t + v_i + \xi_j + \epsilon_{ijt} \quad (7)$$

A vector H_{it} is age dummy variables and a vector of coefficients δ captures the nonlinear relation between mobility and age of head of household. Using less than 60 years old as the reference group, we include five dummy variables in H_{it} . From the estimates of θ , we can see that β will change depending on head of household's age.

Table 4 shows the model for owner households. Model 1 shows that the coefficient of elderly housing with life support service does not give a significant result. However, if we interact the variable with head of household age dummies, some coefficients become significant. Coefficients of the variable that interacted with an indicator of age from 60 to 65 demonstrate significantly negative coefficients irrespective of control groups. This implies that people around 60–65 tend to remain in their housing if they are living in cities where elderly housing with life support service is established.

Table 5 summarized the falsification tests of staggered DID estimation by replacing an indicator of elderly housing with life support service with an indicator of private nursing homes. We observe significantly positive coefficient of indicators of falsified elderly housing with life support service, that is, private nursing homes. People tend to move out from housing and municipalities where there are private nursing homes. Apart from this, we see similar tendencies as in Table 4, in which the coefficients of variable interacting with an indicator of ages 60–65 demonstrating significantly negative and similar magnitudes. Therefore, the results of

¹⁵ We include households headed by those aged 40 years and above because such households start to settle down to their housing and move less and stay more.

Table 4 shows that owners, especially those between 60 and 65, do not move from municipalities with elderly housing with life support service, unlike owners living in municipalities with private nursing homes.

Table 6 demonstrates the results of the renter models. Model 1 shows that the coefficient of elderly housing with life support service dummy does not give significant results. However, the coefficients of interaction terms with the head of household aged from 55 to 60 and 60 to 65 indicators show significantly negative results, particularly in the model with the third control group. Furthermore, in renter models based on the first and third control groups, interaction terms with heads of household aged 75 years and above show negatively significant results.

Table 7 summarizes the falsification tests of staggered DID estimation replacing an indicator of elderly housing with life support service with an indicator of private nursing home. We see similar results as in Table 6, the coefficients of interaction terms with household heads aged 55–60 years and 65–70 years are significantly negative. However, we cannot see significant results of the coefficients of the interaction term with household heads aged 75 years and above, in Table 7. Therefore, renters aged 60–65 years and 75 years and above reduce the probability of moving out from current housing and municipalities, which have established elderly housing with life support services.

In summary, we evaluated the effects of the elderly housing with life support service introduced in November 2011 on residential mobility. We found that owner households headed by people aged 60–65 years stay in their houses and municipalities where the new elderly housing was established. Renters aged 60–65 years and 75 years and above reduced their mobility and tended to live in municipalities where the elderly housing was established.

6. Conclusion

This study used household-level panel and municipal-level data for LTC facilities to examine the mobility of households headed by the elderly aged 60 years and above to test the hypothesis that elderly households have low ability for LTC provision within the households. We selected LTC welfare facilities and private nursing homes as determinants for the relocation of elderly households and analyzed whether such households move out from the municipalities where they live. We estimated several models for both owners and renters with or without children. Furthermore, we evaluated the effects of the elderly housing with life support service introduced in 2011 on residential mobility.

From our analysis, we obtained the following results. Owners do not move out from their housing regardless of the number of beds in LTC or private facilities in the living municipality. Among of them, single female owners without children stay in cities where LTC welfare facilities are accessible. Renters, especially single males without children, tend to stay in cities where private nursing homes are accessible. In the municipality where elderly housing with life support service was introduced, owner and renter households headed by people aged 60–65 years and renters headed by 75 and above reduced their residential mobility.

The results support the hypothesis that we presented in Section 4.2. Owner-occupied elderly do not move out in response to the level of institutional long-term care in their living municipality owing to the difficulty of moving and the high likelihood of at-home care. For elderly owners, the option to move may not be a private nursing home as a substitute for a residence, but an LTC facility where they receive daily care. Therefore, for owner elderly widows without children, when they feel the need for long-term care, their option is to enter a facility rather than at-home care. Conversely, the elderly renter tends to move from a younger age in response to the capacity of LTC or private facilities in the municipality. This is because the renter is more likely to move and less likely to receive at-home care.

Our findings have important implications for public long-term care

and housing policy in Japan. It implies that the necessary care services might differ depending on the type of housing ownership. In Japan, universal long-term care services are provided under the public long-term care insurance system. However, the required long-term care services in a region may vary depending on the ratio of owner-occupied or rented houses. For elderly owner-occupied households, considering how long-term care should be provided to childless households when they need it is necessary. For elderly renter households, considering how to balance between housing and care during old age is crucial. Further, it is difficult to increase the number of LTC facilities because they are designed for elderly people who need severe care and because of the high cost and the problem of securing professional staff. Admission to private nursing homes is also a problem owing to the heavy financial burden on the users. Therefore, housing for the elderly should be developed according to the gradation of their care needs. In this sense, the introduction of elderly housing with life support service can be evaluated positively.

Finally, we remain some ancillary items of note for future studies.

Our results related to the age of household heads suggest that owners and renters are less likely to move as they age. These results based on micro data are in sharp contrast to those of previous studies based on aggregated data that find LTC facilities induce migration of those aged 75 years and above (Nakazawa, 2007; Kawase and Nakazawa, 2009; and Nakazawa and Kawase, 2011). While this current study examines the elderly choosing to move on their own, the aforementioned studies include mobility as a family decision. The number of elderlies who cannot decide on their own will increase in the future. For example, according to the Cabinet Office (Cabinet Office, 2016), it is estimated that the number of elderly people 65 and above with dementia will be about 7 million in 2025, or one in five elderly people. Studying the residential mobility of the elderly, as per family decisions will become more important in the future. Painter and Lee (2009) show that the residential mobility of the elderly relates to the socioeconomic background of their children; however, we could not include this information. Gathering such information is a part of our ongoing research agenda.¹⁶

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Declaration of Competing Interest

None

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¹⁶ Keio University Panel Data Research Center began surveying children of JHPS respondents in 2019.

¹⁶ Keio University Panel Data Research Center began surveying children of JHPS respondents in 2019.

Appendix

A1. Descriptive statistics

See [Tables A 1.](#) and [A 2](#)

Table A1

Descriptive statistics among owner households.

Variable	Whole sample		(a) Mover		(b) Stayer		(a)-(b)
	Mean	S. D.	Mean	S. D.	Mean	S. D.	
move (=1)	0.008	0.087	1.000	0	0	0	
Real housing equity (10,000 JPY)	2,248	3,066	1,741	1,745	2,252	3,074	***
Number of rooms [4 or over] (=1)	0.937	0.242	0.885	0.320	0.938	0.242	*
Number of rooms: missing (=1)	0.008	0.090	0.008	0.087	0.008	0.090	
Condominium (=1)	0.103	0.304	0.229	0.422	0.102	0.303	***
Ground lease (=1)	0.031	0.174	0.061	0.240	0.031	0.173	
Built after 1981 (=1)	0.553	0.497	0.504	0.502	0.554	0.497	
Age of household head	68.0	5.8	67.038	6.341	68.027	5.752	*
Real Income (10,000 JPY)	598	533	648	550	598	533	
Real financial wealth (10,000 JPY)	1,806	2,545	1,969	2,793	1,805	2,543	
Age of household head(/10)	6.802	0.576	6.704	0.634	6.803	0.575	*
College graduate (=1)	0.260	0.439	0.305	0.462	0.260	0.438	
Married (=1)	0.823	0.382	0.702	0.459	0.824	0.381	***
Female household head (=1)	0.188	0.391	0.275	0.448	0.187	0.390	**
Single (=1)	0.093	0.291	0.183	0.388	0.093	0.290	***
Self-rated health	1.654	0.982	1.603	0.934	1.655	0.982	
Full-time worker (=1)	0.127	0.333	0.191	0.394	0.126	0.332	*
Part-time worker (=1)	0.154	0.361	0.198	0.400	0.154	0.361	
Self-employment (=1)	0.181	0.385	0.145	0.353	0.182	0.385	
Retired (=1)	0.271	0.445	0.191	0.394	0.272	0.445	**
Number of family members	2.670	1.255	2.527	1.291	2.671	1.255	
Family decreased since 2004 (=1)	0.131	0.337	0.214	0.412	0.130	0.337	**
Widowhood (=1)	0.054	0.226	0.084	0.278	0.054	0.225	
Living with parents(=1)	0.097	0.296	0.084	0.278	0.097	0.296	
Number of coresident male children	0.294	0.520	0.267	0.461	0.294	0.521	
Number of non-coresident male children	0.206	0.513	0.214	0.569	0.206	0.513	
Number of coresident female children	0.212	0.464	0.244	0.466	0.212	0.464	
Number of non-coresident female children	0.153	0.446	0.145	0.414	0.153	0.446	
No child (=1)	0.234	0.423	0.244	0.431	0.234	0.423	
Difference of log of land rent-price ratio with neighboring prefectures	0.440	0.625	0.571	0.642	0.439	0.625	**
Major cities (=1)	0.257	0.437	0.321	0.469	0.256	0.437	
Other cities (=1)	0.629	0.483	0.565	0.498	0.630	0.483	
N	17,254		131		17,123		

Source: JHPS/KHPS 2004–2018. Notes: Households headed by those aged 60 years and above. The group mean comparison represents the results for a *t*-test of equal means between two samples. This uses Welch's method under the hypothesis of heteroskedasticity. ***, **, and * indicate that estimated coefficients are significant at 1%, 5%, and 10% levels, respectively.

Table A2

Descriptive statistics among renter households.

Variable	Whole sample		(a) Mover		(b) Stayer		(a)-(b)
	Mean	S. D.	Mean	S. D.	Mean	S. D.	
move (=1)	0.053	0.225	1	0	0	0	
Real rent(1000 JPY)	53.6	36.5	70.5	60.2	52.5	34.3	***
Number of rooms [4 or over] (=1)	0.514	0.500	0.552	0.499	0.512	0.500	
Number of rooms: missing (=1)	0.044	0.206	0.056	0.231	0.044	0.205	
Condominium (=1)	0.449	0.498	0.336	0.474	0.456	0.498	***
Wooden apartment (=1)	0.196	0.397	0.232	0.424	0.194	0.395	
Built after 1981 (=1)	0.513	0.500	0.512	0.502	0.513	0.500	
Age of household head	67.611	5.800	66.424	5.229	67.677	5.824	**
Real Income (10,000 JPY)	403.8	392.8	488.8	408.8	399.0	391.5	**
Real financial wealth (10,000 JPY)	590.2	1,310.0	804.5	1,783.8	578.3	1,277.9	
Age of household head(/10)	6.761	0.580	6.642	0.523	6.768	0.582	**
College graduate (=1)	0.171	0.377	0.312	0.465	0.163	0.370	***
Married (=1)	0.582	0.493	0.592	0.493	0.582	0.493	
Female household head (=1)	0.298	0.457	0.296	0.458	0.298	0.458	
Single (=1)	0.316	0.465	0.296	0.458	0.318	0.466	
Self-rated health	1.825	1.026	1.664	0.991	1.834	1.027	*
Self-rated health: missing (=1)	0.010	0.099	0.016	0.126	0.009	0.097	
Full-time worker (=1)	0.134	0.341	0.136	0.344	0.134	0.341	

(continued on next page)

Table A2 (continued)

	Whole sample		(a) Mover		(b) Stayer		(a)-(b)
	Mean	S. D.	Mean	S. D.	Mean	S. D.	
Part-time worker (=1)	0.214	0.410	0.304	0.462	0.209	0.407	**
Self-employment (=1)	0.126	0.332	0.088	0.284	0.128	0.334	
Retired (=1)	0.222	0.416	0.144	0.353	0.226	0.419	**
Number of family members	2.134	1.109	2.320	1.222	2.124	1.101	*
Family decreased since 2004 (=1)	0.138	0.345	0.136	0.344	0.138	0.345	
Widowhood (=1)	0.105	0.307	0.120	0.326	0.104	0.306	
Living with parents(=1)	0.034	0.182	0.048	0.215	0.033	0.180	
Number of coresident male children	0.233	0.498	0.368	0.654	0.226	0.487	**
Number of non-coresident male children	0.175	0.460	0.248	0.549	0.171	0.455	
Number of coresident female children	0.201	0.457	0.208	0.445	0.200	0.458	
Number of non-coresident female children	0.131	0.430	0.120	0.433	0.132	0.430	
No child (=1)	0.347	0.476	0.240	0.429	0.353	0.478	***
Difference of log of land rent-price ratio with neighboring prefectures	0.581	0.678	0.557	0.676	0.582	0.678	
Major cities (=1)	0.369	0.483	0.320	0.468	0.372	0.483	
Other cities (=1)	0.569	0.495	0.584	0.495	0.568	0.495	
N	2,342		125		2,217		

Source: JHPS/KHPS 2004–2015. Notes: Households headed by those aged 60 years and above. The group mean comparison represents the results for a *t*-test of equal means between two samples. This uses Welch's method under the hypothesis of heteroskedasticity. ***, **, and * indicate that estimated coefficients are significant at 1%, 5%, and 10% levels, respectively.

Table A3

Descriptive statistics of measures of the availability of long-term care facilities: Owner households' sample.

	Whole sample		(a) Mover		(b) Stayer		(a)-(b)
	Mean	S.D.	Mean	S.D.	Mean	S.D.	
Average no. of beds of LTC welfare facilities per hundred elderly (0, 20) km	1.330	0.645	1.283	0.648	1.330	0.645	
Average no. of beds of LTC welfare facilities per hundred elderly [20, 50) km	1.504	0.434	1.510	0.414	1.504	0.434	
Average no. of beds of LTC welfare facilities per hundred elderly [50 and above) km	1.568	0.124	1.569	0.130	1.568	0.124	
No. of beds of LTC welfare facilities per hundred elderly in current municipality	1.335	0.682	1.247	0.693	1.336	0.682	
Average of private nursing homes per hundred elderly (0, 20) km	0.753	0.585	0.881	0.600	0.752	0.585	**
Average no. of beds of private nursing homes per hundred elderly [20, 50) km	0.650	0.408	0.718	0.401	0.649	0.408	*
Average no. of beds of private nursing homes per hundred elderly [50 and above) km	0.519	0.201	0.527	0.183	0.519	0.201	
No. of beds of private nursing homes per hundred elderly in current municipality	0.978	0.893	1.089	0.906	0.977	0.893	
N	17,254		131		17,123		

Note: "km" stands for the kilometers. (a)-(b) means the group mean comparison of the results for a *t*-test of equal means between two samples. This uses Welch's method under the hypothesis of heteroskedasticity.

Table A4

Descriptive statistics of measures of the availability of long-term care facilities: Renter households' sample.

	Whole sample		(a) Mover		(b) Stayer		(a)-(b)
	Mean	S.D.	Mean	S.D.	Mean	S.D.	
Average no. of beds of LTC welfare facilities per hundred elderly (0, 20) km	1.217	0.477	1.264	0.603	1.214	0.469	
Average no. of beds of LTC welfare facilities per hundred elderly [20, 50) km	1.530	0.417	1.550	0.418	1.529	0.417	
Average no. of beds of LTC welfare facilities per hundred elderly [50 and above) km	1.575	0.127	1.580	0.140	1.575	0.126	
No. of beds of LTC welfare facilities per hundred elderly in current municipality	1.207	0.565	1.161	0.546	1.210	0.566	
Average of private nursing homes per hundred elderly (0, 20) km	0.844	0.591	0.794	0.586	0.846	0.592	
Average no. of beds of Private nursing homes per hundred elderly [20, 50) km	0.684	0.410	0.662	0.410	0.686	0.410	
Average no. of beds of Private nursing homes per hundred elderly [50 and above) km	0.515	0.199	0.500	0.199	0.516	0.199	
No. of beds of private nursing homes per hundred elderly in current municipality	1.127	0.924	0.979	0.864	1.136	0.926	*
N	2,342		125		2,217		

Note: "km" stands for the kilometers. (a)-(b) means the group mean comparison of the results for a *t*-test of equal means between two samples. This uses Welch's method under the hypothesis of heteroskedasticity.

A2. Measures of the availability of long-term care facilities

For the amenity of the neighboring municipality A_{ih} in Eq. (3), we calculate the average numbers in neighboring municipalities where the household could move to. We need to calculate the number of beds per elderly in LTC facilities at a new location h where the elderly households would potentially be able to move to.

First, we define the number of facility beds per elderly (*NBPE*) in the present municipality as follows (Nakazawa, 2007; Nakazawa and Kawase, 2011):

$$NBPE_{ij} = \frac{NB_{ij}}{POP65OVER_{ij}}, \quad (8)$$

where NB_{ij} is the number of facility beds in a municipality j where household i lives and $POP65OVER_{ij}$ is the population over 65 years old in the municipality j where household i lives.

Thereafter, the average number of elderly facility beds in neighboring municipalities within a d -km radius from the municipalities where the elderly household is located is calculated as follows:

$$\overline{NBPE}^d = W_d \times NBPE,$$

and the i th observation is expressed as follows:

$$\overline{NBPE}_i^d = \sum_{j=1}^n w_{dij} NBPE_{ij}. \quad (9)$$

Here, W_d is the spatial weight matrix, and the (i,j) th element of this w_{dij} is the inverse of the distances between the central point¹⁷ of municipality where household i lives and the central point of another municipality j within a d -km radius, where $d < 20$, $20 \leq d < 50$, and $d \geq 50$ km. In particular, $d < 20$ km is for mobility within the municipality or neighboring municipalities, $20 \leq d < 50$ km for mobility within the prefecture or neighboring prefectures, and $d \geq 50$ km for mobility to other prefectures within the eight regions. All units that are observed within a d -km radius are assumed to obtain the j -th row of W_d . In general, an inverse correlation exists between municipalities' distance from the present residence and the likelihood of moving there. Considering this, the spatial weights matrix W_d is based on the inverse of the distance between municipalities. In addition, we used other spatial weight matrices based on a dummy variable that takes the value of 1 if the present municipality j is observed within a d -km radius, and 0 otherwise. We also used the spatial weight matrix indicating the neighboring prefecture where household i in city j is located. The elements of the spatial weight matrix are row normalized.

We estimated the model of Eq. (3) using $NBPE_{ij}$ and $\overline{NBPE}_i^d$ measured at time t for A_{ijt} and A_{iht} as the baseline model. These measures are calculated for LTC facilities, private nursing homes. Descriptive statistics of these variables are tabulated in Table A3 for owners and Table A4 for renters, respectively.

We obtained population data from the "Report on Internal Migration in Japan" by the Statistics Bureau (2003–2017), which provides basic resident registration information for each municipality. For LTC facilities, we consulted the Ministry of Health, Labour, and Welfare (2003–2017a) "Survey of Institutions and Establishments for Long-term Care." We derived the number of private nursing home beds from the Ministry of Health, Labour, and Welfare (2003–2017b) "Survey of Social Welfare Institutions." Finally, we draw the number of beds of hospitals and clinics from the Ministry of Health, Labour, and Welfare (2003–2017c) "Survey of Medical Institutions."

A3. Robustness checks

As written in Section 5, the baseline models do not consider several points. For robustness checks, we estimate several models to consider the following three points: (i) estimation considering unobserved heterogeneity, (ii) estimation considering rare event nature of outcome, and (iii) estimation with alternative modeling approaches.

First, as shown by previous literature, residential mobility of elderly households is affected by many factors. We could not observe some factors, such as preference for the living place of the households reflecting family history living in the same place over several generations, and bequest motive of housing to their heirs, etc. If such unobserved heterogeneities were correlated with explanatory variables, these would become a source of the bias of the estimated coefficients. Therefore, we should consider that to control the unobserved heterogeneity of elderly households.

To control the unobserved heterogeneity, we estimate linear probability models with fixed effects. The necessity of a household's fixed effects in the models is supported from many specification tests of linear models (against pooled OLS and random effects models). However, some coefficients of household characteristics of those estimation results, such as college degree, show counterintuitive results. This is because those variables rarely change or are slow changing. For example, some elderly go to college after their retirement; however, such people are few. Average number of facility beds are another example of slow changing variables. Although we can estimate the coefficients of rarely changing variables, the standard errors become large and the point estimate of them becomes unreliable (Plumper and Troeger, 2007). To remedy this problem, some methods are proposed in the literature, such as random effects model as formulated by Mundlak (1978), the fixed effects vector decomposition method (Plumper and Troeger, 2007), and within-between formulation (Bell and Jones, 2015). These models have different characteristics and should be selected for the specific purpose of the analysis; however, they require estimating a lot of parameters and it is almost impossible to estimate them due to the rare nature of elderly household's mobility. After estimating several models for preliminary analysis, we selected pooled OLS models as our baseline model and checked the robustness by estimation of other models.

Second, as we observed in Section 4, residential mobility is a very rare event. According to the previous literature (King and Zeng, 2001; for the survey, Maalouf, 2011), parameter estimates of the logit model based on rare events data has a bias that underestimates the probability of an event occurring, $\text{Prob}(y = 1)$, and overestimates that of a non-occurring event, $\text{Prob}(y = 0)$.¹⁸ Therefore, we employ the Firth (1993) logit model to reduce the bias, estimate the binary choice model, and explain residential mobility. However, in this estimation, we cannot control the unobserved heterogeneity as we do not have enough of a mover sample to estimate the fixed effect logit model (Chamberlain, 1980). Therefore, we estimated Firth's logit models to see the bias due to rare event mobility.

Third, to check the robustness of our baseline results, we estimated survival models taking advantage of panel data as an alternative approach. In these models, hazard rate of the households to move out from the residence is estimated by using the same covariates used in baseline models. By

¹⁷ The central point of the municipality is where the city hall is located.

¹⁸ Events per explanatory variable (EPV) is used to show the reliabilities of the estimated logit models and an EPV less than 10 should be interpreted with caution because some biases are included in the estimates in the positive or negative directions, as per the epidemiology literature (Peduzzi et al., 1996; Ottenbacher et al., 2004).

Table A5

Estimation results of the alternative models of mover households' linear probability model with fixed effects model.

Facility/variables	1. Owner	(b)Private nursing	2. Renter	(b)Private nursing
	(a) LTC welfare facility	home	(a) LTC welfare facility	home
Average no. of beds of facilities per hundred elderly (0, 20) km, ln	−0.00649 (0.00921)	0.00499 (0.00616)	0.00401 (0.0561)	0.0626 (0.0511)
Average no. of beds of facilities per hundred elderly [20, 50) km, ln	0.0149 (0.0162)	0.0159* (0.00927)	−0.124 (0.0968)	0.00786 (0.0686)
Average no. of beds of facilities per hundred elderly [50 and above) km, ln	0.197 (0.234)	−0.0579 (0.0448)	−0.943* (0.550)	0.284 (0.406)
No. of beds of facilities per hundred elderly, ln	−0.00287 (0.00478)	0.000839 (0.00523)	−0.0484 (0.0347)	−0.0441 (0.0283)
Number of observations	17,254	17,254	2342	2342
R-squared	0.030	0.030	0.075	0.073
Number of events	131	131	125	125
Number of variables	14,078	14,078	1823	1823
EPV	0.00931	0.00931	0.0686	0.0686
Number of households	3,177	3,177	520	520
Hausman test: FE vs. RE	422.3	425.4	181	167.6
Hausman test P values	0.000	0.000	0.000	0.000
F test for fixed effects	2.254	2.252	1.604	1.586
F test P value	0.000	0.000	0.000	0.000

Note: Cluster standard errors over households are in parentheses. Significance level: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. “km” and “ln” stand for the kilometers and natural logarithm, respectively. The same covariates and fixed effects as Table 2 are used, but their coefficients are suppressed.

analyzing the estimated coefficients of those models, we can verify the robustness of our benchmark models qualitatively.

In summary, we estimated the linear probability models of elderly household's mobility by OLS as benchmark models. For robustness checks, we compared the results with those of the fixed effects models, rare events logit models, and survival models.

A3.1. Fixed effects models

If unobserved heterogeneities, which are related to the households, had correlations with the explanatory variables, these would become sources of the bias of the estimated coefficients. Table A5 shows the estimation results of owner's residential mobility model with fixed effects. These fixed effects models are selected through specification tests: F test for fixed effects or pooled OLS models and Hausman tests for fixed effects or random effects models. In owner models, coefficients of average number of beds of LTC welfare facilities in neighboring municipality show insignificant results, which are similar to Table 2, but signs of the coefficients of private nursing homes [20, 50) are changed from Table 2 and become significant. Meanwhile, in renter models, the coefficients of the average number of beds of LTC facilities in neighboring municipality become significant. These signs and magnitude of the coefficients are changed from Table 2. In terms of clustered standard errors, they become larger than those in Table A5. We believe that those changes from Table 2 are caused by the low within variation of those variables due to slow change over time and the within transformation of the fixed effects models, and rare event bias.

For example, if the model is specified as follows:

$$y_{it} = \gamma x'_{it} + \nu_i + u_{it},$$

where x'_{it} is a vector of $1 \times K$, ν_i is a fixed effect, and u_{it} is a stochastic error term. The average of variables over time provides

$$\bar{y}_i = \gamma \bar{x}_i + \nu_i + \bar{u}_i.$$

Subtracting this equation from the above equation obtains

$$y_{it} - \bar{y}_i = \gamma (x_{it} - \bar{x}_i) + (u_{it} - \bar{u}_i) = \gamma (x_{it} - \bar{x}_i) + \epsilon_{it},$$

where $\epsilon_{it} = u_{it} - \bar{u}_i$ is iid without loss of generality, after the within transformation, $\hat{\gamma}$, which is a coefficient of covariates x'_{it} is calculated as follows:

$$\hat{\gamma} = \left(\sum_i \sum_t (x_{it} - \bar{x}_i) (x_{it} - \bar{x}_i)' \right)^{-1} \sum_i \sum_t (x_{it} - \bar{x}_i) (y_{it} - \bar{y}_i).$$

Through the within transformation, the slow changing variables are not wiped out. For example, if x_{it} is an average number of beds of facilities in neighboring municipalities, its mean $\bar{x}_i$ is an average of the averages and its variance becomes smaller as the number of neighboring municipalities becomes larger. Therefore, the difference from its average $x_{it} - \bar{x}_i$ can be positive or negative with a small difference and can have various signs from the estimates of the pooled OLS in Table 2.

Meanwhile, asymptotic variance of $\hat{\gamma}$ is calculated from the following equation:

$$\text{Var}(\hat{\gamma}) = \hat{\sigma}_{\epsilon}^2 \left(\sum_i \sum_t (x_{it} - \bar{x}_i) (x_{it} - \bar{x}_i)' \right)^{-1}.$$

Through the within transformation on a variable with little within variance due to the slow change of x_{it} over time, the denominator becomes small. The coefficients of the average number of beds of facilities in neighboring municipalities are the typical example. The variance of the estimates

of the FE estimator can approach infinity and become unreliable (Plumper and Troeger, 2007). We believe that this leads to larger standard errors in Table A5 than those in Table 2.

In addition to these problems, fixed effects estimates can be suffering from rare event bias. We need to pay attention to the bias if the number of events per explanatory variable, EPV, is less than 10 according to epidemiology literature (Peduzzi et al., 1996; Ottenbacher et al., 2004). EPVs of these models in Table A5 are much lower than 10. Therefore, these estimates can have rare event biases.

Thus, we employed pooled OLS models without household fixed effects as baseline models to avoid the problems we encounter when we estimate the fixed effects models to explain rare events from the data including slow changing variables.

A3.2. Logit models

Table A6 shows the estimation results of logit models. Each model is estimated by the logit model and Firth's logit model. Despite the low EPV value of about 2, the logit model and Firth's logit model provided similar results. This indicates that the bias due to the rare case of residential mobility is small. Furthermore, the marginal effects of probability to move show similar effects with linear probability models of owners in Table 2.

The potential criticism about the spatial models in Table A6 is the selection of the spatial matrix. Thus, we estimate the models with other spatial matrices for robustness checks. Table A7 summarizes the estimation results based on alternative spatial weight matrices. The models in the table are based on alternative weight matrices from $w_{ij} = 1$ if the unit i is observed within a d -km radius ($d < 20$, $20 \leq d < 50$, and $d \geq 50$ km), and 0 otherwise. The average partial effects on probability to move show similar results with Table A6.

Our model includes spatial terms on the righthand side. This model is the spatial Durbin model (LeSage and Pace, 2009). Elhorst (2010) suggested that an effective criterion for selecting a weight matrix is comparing maximum log-likelihood function value of spatial models, as proposed by Stakhovych and Bijmolt (2008). Following these studies, we selected the model from the estimation results of logit models. By comparing the estimated models of owners in Table A6, which uses weight matrices from $w_{ij} = 1/d_{ij}$ if the unit i is observed within a d -km radius, and 0 otherwise, we can observe that log-likelihood function values are almost the same as those of specifications in Table A7.

A3.3. Survival models

Table A8 demonstrates the estimation results of survival models. These models are estimated by using the same dataset. However, the years of residential spell in the same house is explained in the survival analysis. Cox's proportional hazard models are estimated. The interesting coefficients of the models are tabulated. The number of the observations is smaller compared with those of Table 2. This is due to the observations including missing cases of residential spell that are not used for the previous analysis. In renter's model, the coefficients of average number of LTC facilities show no significant results, which depict similar results to Table 2. Renter households stay in their houses if the average number of beds of the LTC welfare facility increases.

Table A6
Estimation results of the logit models of mover households.

Facility variable	(a) LTC welfare facilities		(b) Private nursing homes	
	logit	Firth logit	logit	Firth logit
1. Owner households				
Average no. of beds of facilities per hundred elderly (0, 20) km, ln	-0.00204 (0.00394)	-0.00235 (0.00384)	0.00358 (0.00363)	0.00436 (0.00360)
Average no. of beds of facilities per hundred elderly [20, 50) km, ln	0.00114 (0.00552)	0.00187 (0.00543)	-0.00457 (0.00513)	-0.00572 (0.00511)
Average no. of beds of facilities per hundred elderly [50 and above) km, ln	0.00778 (0.0327)	0.0143 (0.0309)	0.0103 (0.0269)	0.0124 (0.0267)
No. of beds of facilities per hundred elderly, ln	-0.00213 (0.00260)	-0.00255 (0.00260)	0.00136 (0.00194)	0.00162 (0.00198)
Number of observations	17,254	17,254	17,254	17,254
Number of events	131	131	131	131
Number of variables	61	61	61	61
EPV	2.148	2.148	2.148	2.148
Maximum log-likelihood	-719.5	-641.8	-719	-640.7
2. Renter households				
Average no. of beds of facilities per hundred elderly (0,20) km, ln	0.0234 (0.0235)	0.0276 (0.0233)	0.0137 (0.0254)	0.0156 (0.0257)
Average no. of beds of facilities per hundred elderly [20, 50) km, ln	0.0174 (0.0508)	0.0227 (0.0492)	0.00510 (0.0333)	0.00484 (0.0344)
Average no. of beds of facilities per hundred elderly [50 and above) km, ln	0.0474 (0.281)	0.0774 (0.272)	0.156 (0.199)	0.170 (0.197)
No. of beds of facilities per hundred elderly, ln	-0.0409** (0.0182)	-0.0455** (0.0182)	-0.0361** (0.0170)	-0.0401** (0.0170)
Number of observations	2,342	2,342	2,342	2,342
Number of events	125	125	125	125
Number of variables	61	61	61	61
EPV	2.049	2.049	2.049	2.049
Maximum log-likelihood	-436.3	-365.8	-435.5	-364.1

Note: Cluster standard errors over households are in parentheses. Significance level: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. "km" and "ln" stand for the kilometers and natural logarithm, respectively. The same covariates and fixed effects as Table 2 are used but their coefficients are suppressed.

Table A7

Average partial effects of logit models with alternative spatial weights.

Facility variable	(a) LTC welfare facilities		(b) Private nursing homes	
	logit	Firth logit	logit	Firth logit
1. Owner households				
Average no. of beds of facilities per hundred elderly (0,20) km, ln	−0.00150 (0.00447)	−0.00153 (0.00431)	0.00538 (0.00434)	0.00660 (0.00426)
Average no. of beds of facilities per hundred elderly [20, 50) km, ln	0.000245 (0.00824)	0.000400 (0.00805)	−0.00714 (0.00658)	−0.00854 (0.00657)
Average no. of beds of facilities per hundred elderly [50 and above) km, ln	−0.120 (0.255)	−0.141 (0.255)	−0.0643 (0.145)	−0.0663 (0.145)
No. of beds of facilities per hundred elderly, ln	−0.00267 (0.00249)	−0.00323 (0.00248)	0.000951 (0.00199)	0.00114 (0.00203)
Number of observations	17,254	17,254	17,254	17,254
Number of events	131	131	131	131
Number of variables	61	61	61	61
EPV	2.148	2.148	2.148	2.148
Maximum log-likelihood	−719.4	−644	−718.3	−641.8
2. Renter households				
Average no. of beds of facilities per hundred elderly (0, 20) km, ln	0.0225 (0.0308)	0.0273 (0.0305)	0.0429 (0.0307)	0.0486 (0.0305)
Average no. of beds of facilities per hundred elderly [20, 50) km, ln	0.0213 (0.0646)	0.0246 (0.0639)	0.0631 (0.0424)	0.0694 (0.0431)
Average no. of beds of facilities per hundred elderly [50 and above) km, ln	0.567 (1.891)	0.616 (1.940)	2.707** (1.164)	3.039*** (1.157)
No. of beds of facilities per hundred elderly, ln	−0.0398** (0.0183)	−0.0445** (0.0183)	−0.0333** (0.0161)	−0.0369** (0.0162)
Number of observations	2,342	2,342	2,342	2,342
Number of events	125	125	125	125
Number of variables	61	61	61	61
EPV	2.049	2.049	2.049	2.049
Maximum log-likelihood	−436.5	−368.3	−433.2	−363.8

Note: Cluster standard errors over households are in parentheses. Significance level: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. “km” and “ln” stand for the kilometers and natural logarithm, respectively. The same covariates and fixed effects as Table 2 are used but their coefficients are suppressed.

Table A8

Alternative estimation of owner's model: survival models.

Facility/ variables	(1) Owner (a) LTC welfare facility	(b) Private nursing home	(2) Renter (a) LTC welfare facility	(b) Private nursing home
Average no. of beds of facilities per hundred elderly (0, 20) km, ln	−0.144 (0.538)	0.434 (0.474)	−0.0370 (0.523)	0.495 (0.616)
Average no. of beds of facilities per hundred elderly [20, 50) km, ln	−0.634 (0.710)	−0.306 (0.677)	0.221 (1.007)	−0.0435 (0.907)
Average no. of beds of facilities per hundred elderly [50 and above) km, ln	−1.045 (5.131)	1.701 (3.687)	3.163 (6.942)	−1.793 (4.484)
No. of beds of facilities per hundred elderly, ln	0.00879 (0.399)	0.144 (0.261)	−0.828** (0.365)	−0.617 (0.389)
Number of observations	16,640	16,640	2,048	2,048
Number of events	117	117	87	87
Number of variables	59	59	59	59
EPV	1.983	1.983	1.475	1.475
Maximum log-likelihood	−656.3	−655.9	−312	−312.1

Note: Cluster standard errors over households are in parentheses. Significance level: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. “km” and “ln” stand for the kilometers and natural logarithm, respectively. The same covariates and fixed effects as Table 2 are used, but their coefficients are suppressed.

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